

Comprehensive Analysis and Improvement of Species Classification in Seafood

1. Executive Summary :

This project focuses on enhancing the accuracy of species classification for seafood using deep learning techniques. The goal is to develop a robust model that minimizes misclassifications in a diverse dataset containing 46 species, providing a roadmap for future improvements and refinements.

2. High-Level Tech Stack:

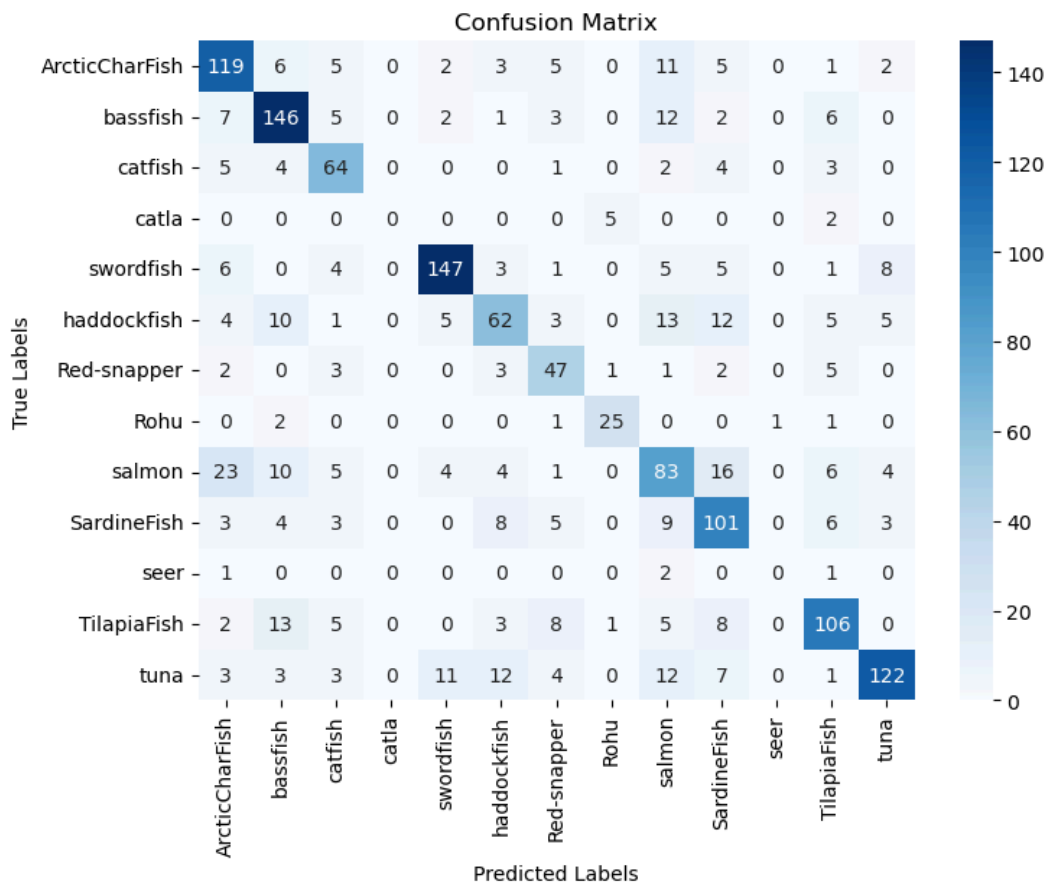
Common Image processing libraries, Frameworks and Development platforms used for all the classification models :

- **Deep Learning Framework:** Tensorflow with Keras, Pytorch
- **Programming:** Python
- **Image processing libraries:** OpenCV, Pillow, Ultralytics, Scikit-Image
- **Data visualization and Plotting:** Matplotlib, Seaborn
- **Development platform:** Jupyter notebook, Spyder

3. Project Background:

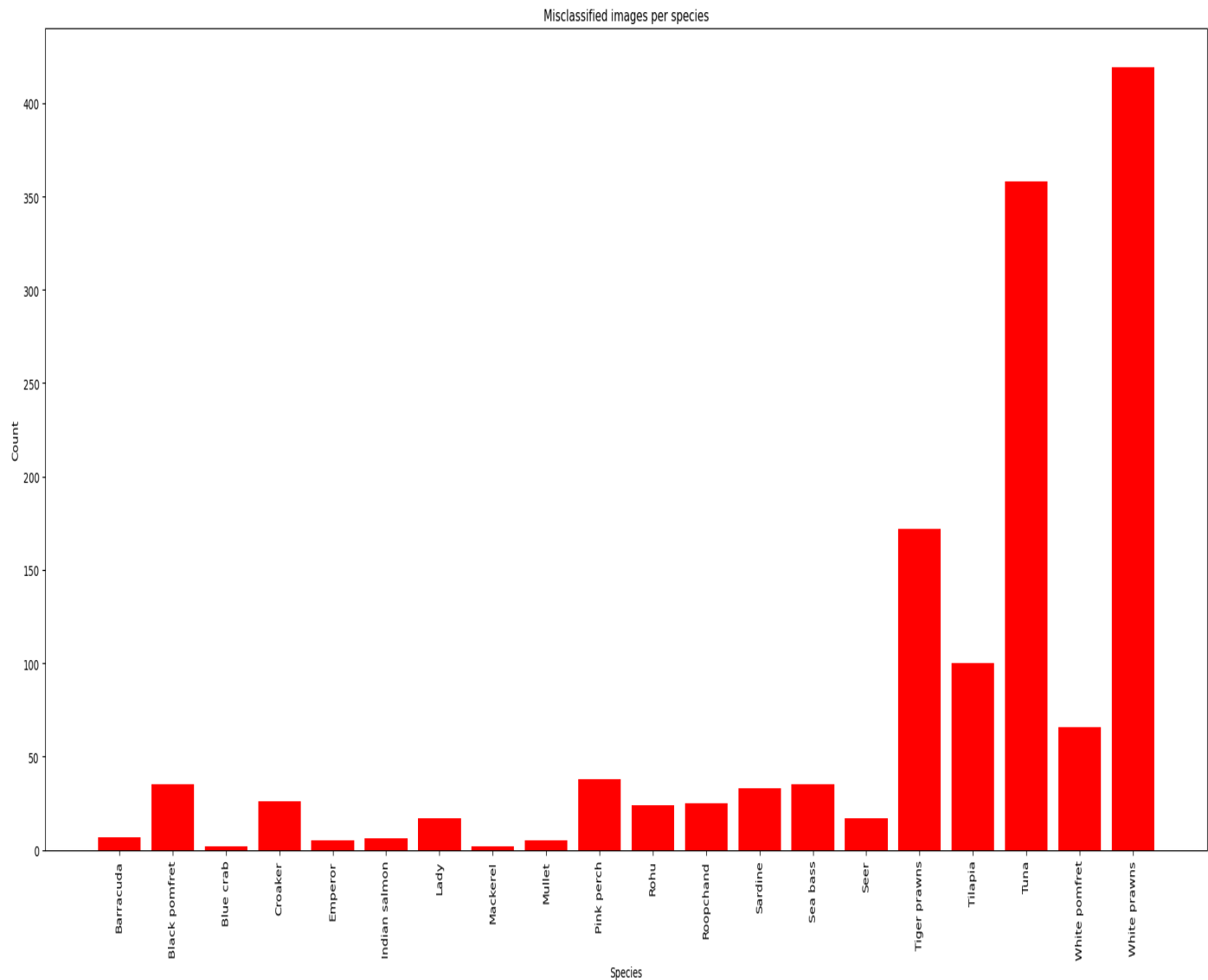
- Initial species classification project with **Kaggle** dataset achieved **70%** accuracy.
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	precision	recall	f1-score	support
ArcticCharFish	0.68	0.75	0.71	159
Red-snapper	0.59	0.73	0.66	64
Rohu	0.78	0.83	0.81	30
SardineFish	0.62	0.71	0.66	142
TilapiaFish	0.74	0.70	0.72	151
bassfish	0.74	0.79	0.76	184
catfish	0.65	0.77	0.71	83
catla	1.00	0.00	0.00	7
haddockfish	0.63	0.52	0.57	120
salmon	0.54	0.53	0.53	156
seer	0.00	0.00	0.00	4
swordfish	0.86	0.82	0.84	180
tuna	0.85	0.69	0.76	178
accuracy			0.70	1458
macro avg	0.67	0.60	0.59	1458
weighted avg	0.71	0.70	0.70	1458



➤ **Model Architecture used :**

- MobileNetV2(all layers freezed)
 - Global Average Pooling 2D
 - 256 neurons dense layer
 - 180 neurons dense layer
 - 0.5 dropout rate
 - 50 neurons dense layer
 - 0.5 dropout rate
 - Softmax activation function
- Adam optimizer, sparse categorical cross entropy, 20 epochs, default learning rate
- Subsequent focus on freshness classification (mackerel and sardine).
- Return to species classification due to misclassifications in the warehouse dataset (more than **1000** misclassified images).

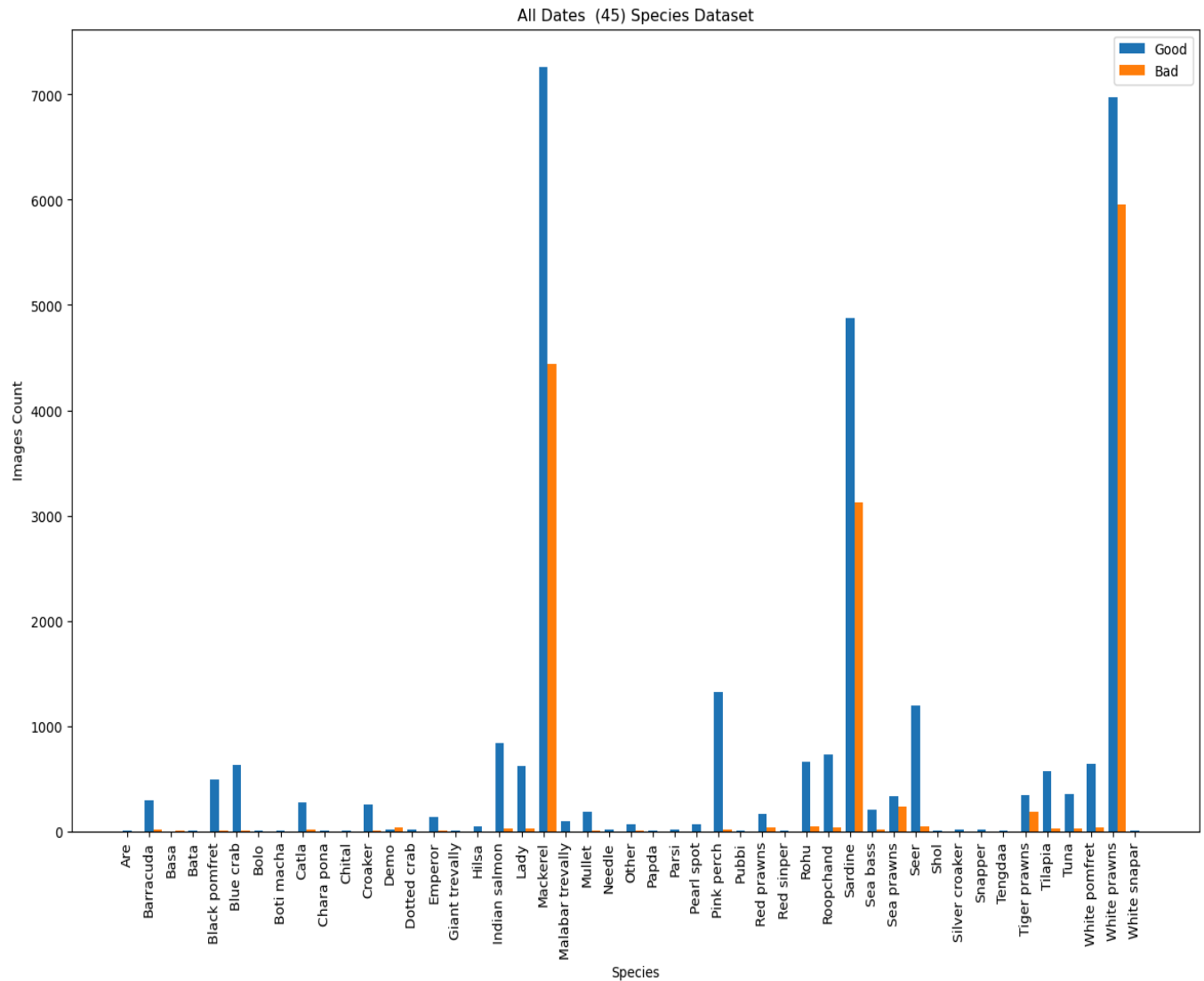


4. Dataset Analysis:

3.1 Data Collection and Distribution(till Feb 05):

- The dataset, as of February 5, 2024, consists of 45 species with an uneven distribution.
- Some species have fewer than 100 images, yet misclassifications persist.
- Total images Count: 45554
- Total Good images Count: 29762

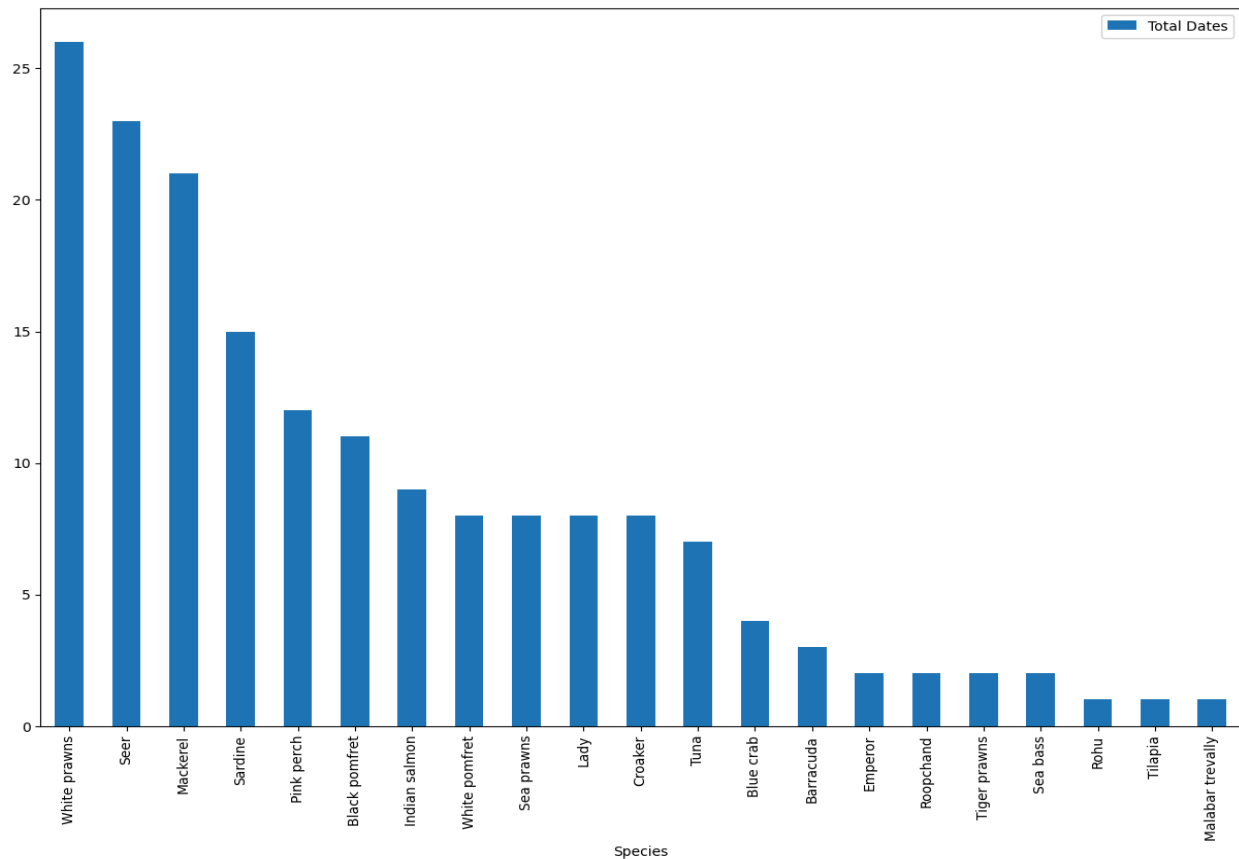
- Total Bad images Count: 14362
- Total Misclassified images Count: 1430
- Total trainable images Count: 44107
- Total Number of Species: 45



3.2 Train Dataset Overview:

- Focused on 24 species, emphasizing those collected in January 2024, species with misclassified images, and those with over 100 images.

Jan Data Collection



5. Performance Metrics:

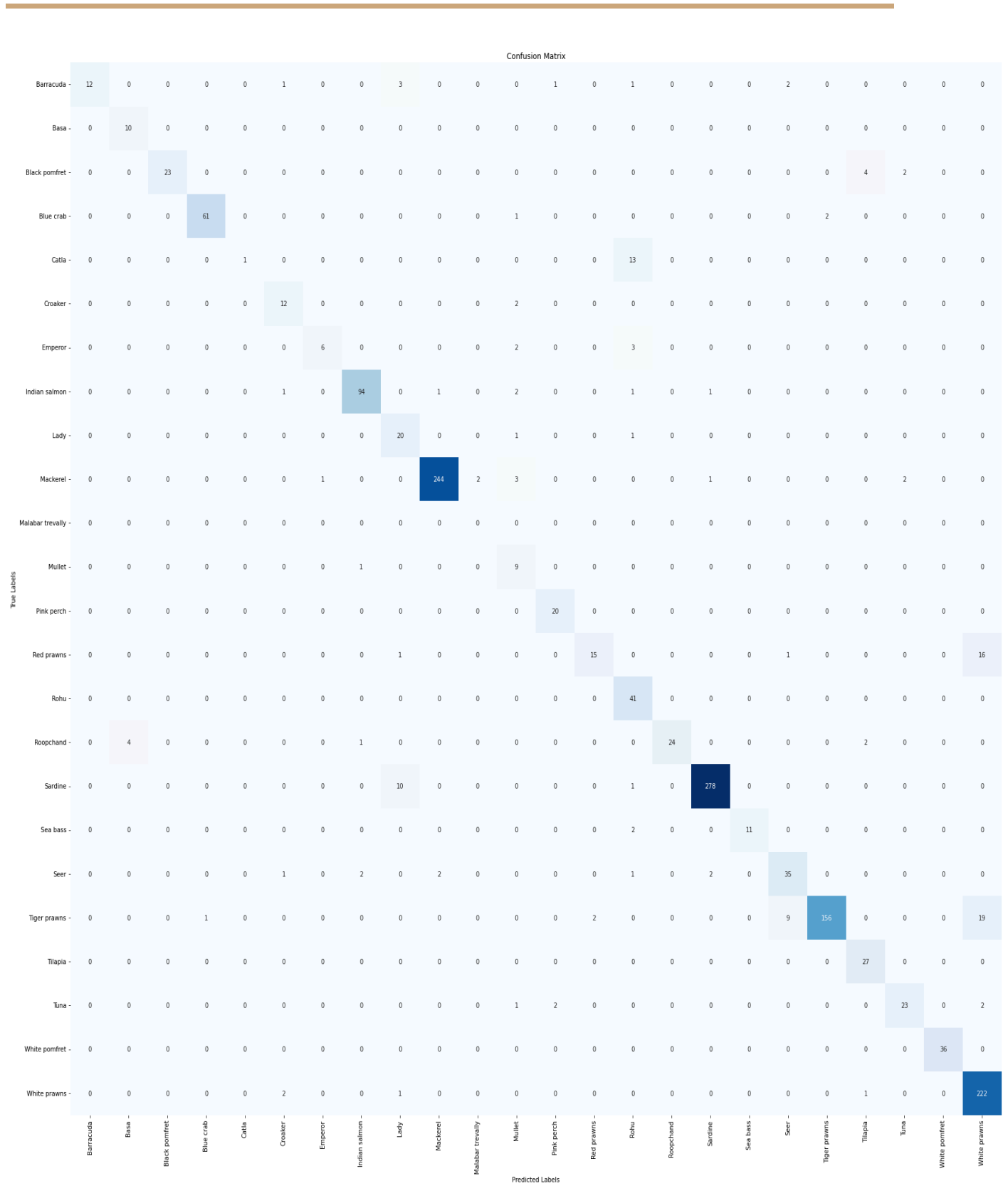
- **Accuracy** gives the overall correctness of the predictions.
 - $\text{Accuracy} = (\text{Correct predictions}) / (\text{Total predictions})$
- **Precision** measures how many of the predicted positive cases were actually positive.
 - $\text{Precision} = (\text{True Positives}) / (\text{True Positives} + \text{False Positives})$
- **Recall** measures how many of the actual positive cases were correctly predicted.
 - $\text{Recall} = (\text{True Positives}) / (\text{True Positives} + \text{False Negatives})$

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- **F1-score** is the harmonic mean of precision and recall and provides a balance between them.
 - $F1\text{-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

6. Model versions:

5.1 Common model for all species

- Training the Model:
 - Trained the model using an imbalanced dataset including fish, crabs and prawns, achieving an overall accuracy of 91%.
 - Utilized MobileNetV2 architecture with sparse categorical loss and configured with 40, 50, and 50 neurons
- Challenges:
 - Catla, Rohu got misclassified and most of the prawns were predicted as white prawns.
 - Some Tiger prawns were predicted as seer.
 - Some Sardine were predicted as lady.



4.2 Two-Stage Strategy

➤ Approach:

- Initial classification into fish, prawn, and crab classes.
- Further species classification within each class.

4.2.1 Model Development: Fish, Prawn, Crab Classification

➤ Initial Model Training:

- Trained with 400 images each for fish, prawns, and crabs.
- Used DenseNet architecture, 60 and 80 neurons, 0.2 dropout, and softmax activation.
- Macro average F1 score loss and 5 epochs.
- Achieved good overall accuracy, but some misclassifications persisted.

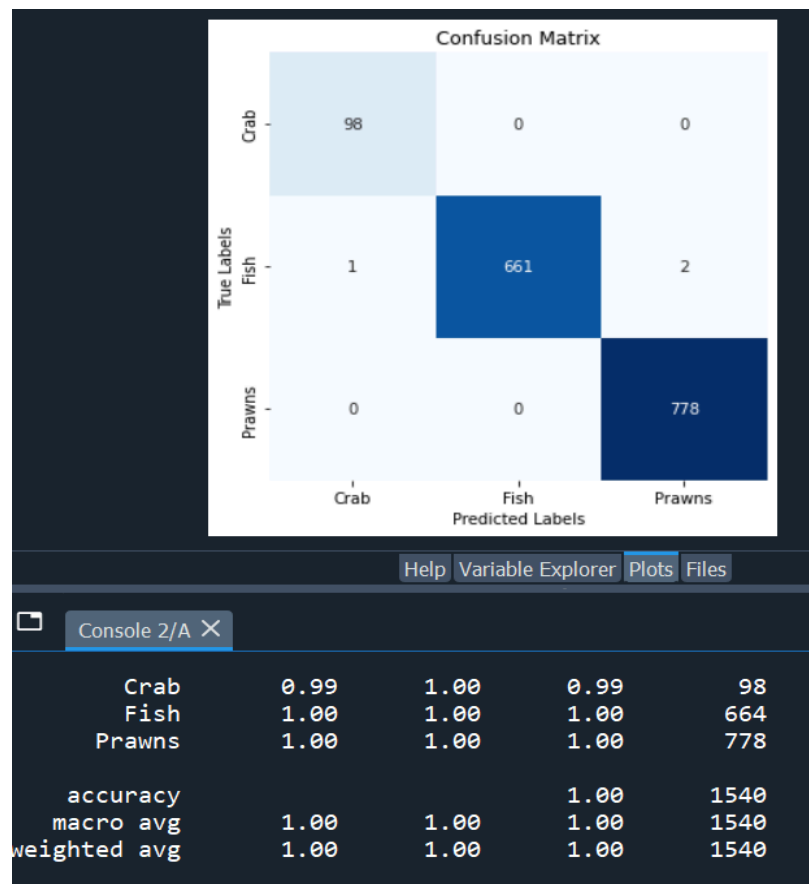


➤ **Segmentation Consideration:**

- Recognized the absence of a segmentation model for crabs.
- Deferred segmentation experiments for crabs due to the current limitations.

➤ **Refinement:**

- Increased crab dataset to 500, fish, and prawns to 1000 each.
- Used the same architecture and loss with 5 epochs.
- Improved results with increased dataset size with 100% accuracy.
- Achieved promising results with accurate predictions for most species.



➤ **Model Testing Results:**

- Tested the model with the entire dataset (up to February 12, 2024).
- High accuracy observed, with only a few misclassifications:
- **37** white prawns misclassified out of **12,925** images.
- **4** seer predicted as prawns out of **1,242**.

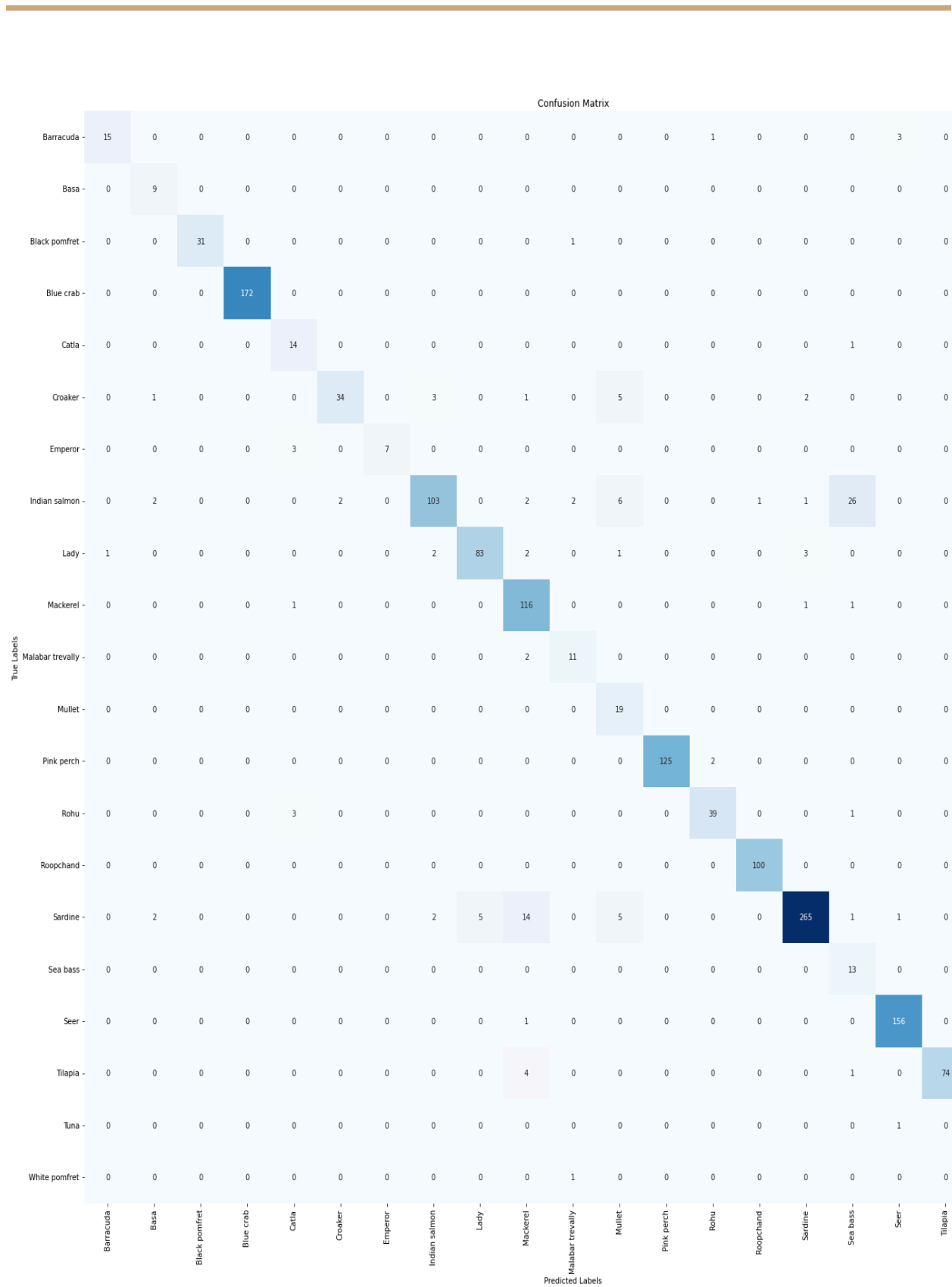
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- **20** sardines were misclassified out of **7,991**.
 - **7** mackerel predicted as prawns out of **11,703**.
 - Other species classified accurately into fish, prawns, and crabs categories.

4.2.2 Model Development: Fish Species Classification

➤ Version1

- **Train Dataset:** Barracuda - 286, Basa - 275, Black pomfret - 439, Blue crab - 456, Catla - 277, Croaker - 195, Emperor - 118, Indian salmon - 639, Lady - 514, Mackerel - 665, Malabar trevally - 83, Mullet - 151, Pink perch - 784, Rohu - 661, Roopchand - 614, Sardine - 748, Sea bass - 207, Seer - 712, Tilapia - 488, Tuna - 335, White pomfret - 616
- **Model Training:** Trained with an imbalanced dataset for 21 fish species for 5 epochs.
- **Model Architecture:** MobileNetV2 as base model, Flatten, 40, 50 and 50 neurons, softmax function and macro average F1 score loss.
- **Test results:** Initially achieved **90%** accuracy, with misclassifications of Indian salmon, sea bass, and occasional sardine misclassifications. The macro average f1-score is **84%**.

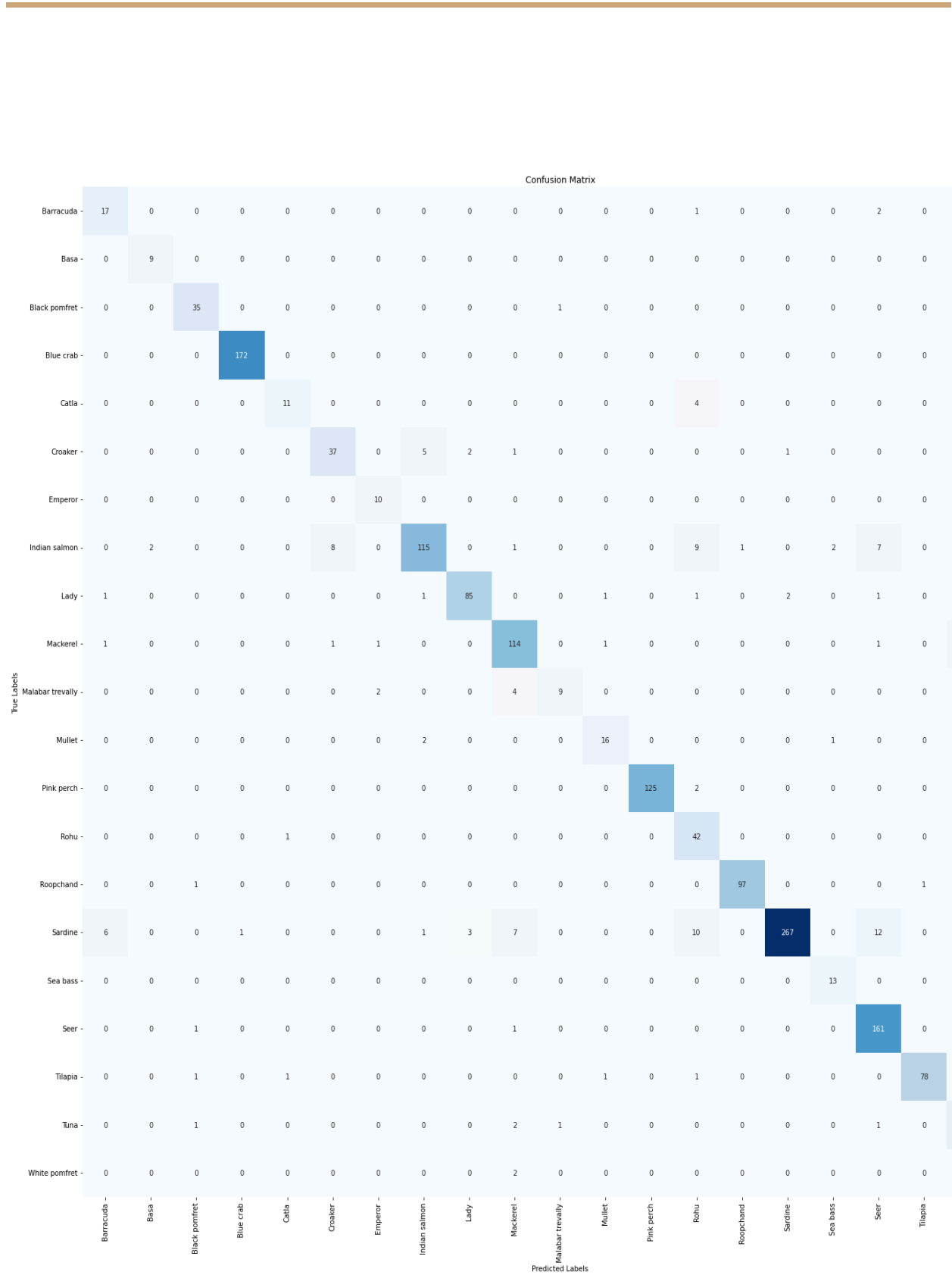
	precision	recall	f1-score	support
Barracuda	0.94	0.75	0.83	20
Basa	0.64	0.90	0.75	10
Black pomfret	1.00	0.84	0.91	37
Blue crab	1.00	1.00	1.00	172
Catla	0.67	0.93	0.78	15
Croaker	0.94	0.74	0.83	46
Emperor	1.00	0.70	0.82	10
Indian salmon	0.94	0.71	0.81	145
Lady	0.94	0.90	0.92	92
Mackerel	0.82	0.91	0.86	128
Malabar trevally	0.73	0.69	0.71	16
Mullet	0.53	1.00	0.69	19
Pink perch	1.00	0.98	0.99	127
Rohu	0.93	0.91	0.92	43
Roopchand	0.99	1.00	1.00	100
Sardine	0.97	0.86	0.92	307
Sea bass	0.30	1.00	0.46	13
Seer	0.97	0.96	0.96	163
Tilapia	1.00	0.90	0.95	82
Tuna	0.44	0.96	0.61	28
White pomfret	0.85	0.97	0.91	36
accuracy			0.90	1609
macro avg	0.84	0.89	0.84	1609
weighted avg	0.93	0.90	0.91	1609



➤ **Version2**

- **Model Training:** Increased epochs to 10 with the same architecture and loss, resulting in improved recall for all classes.
- **Test Results:** The overall accuracy is **91%** but **3%** increase in macro average f1-score(**87%**). Indian salmon and Malabar trevally has less recall.

	precision	recall	f1-score	support
Barracuda	0.68	0.85	0.76	20
Basa	0.82	0.90	0.86	10
Black pomfret	0.90	0.95	0.92	37
Blue crab	0.99	1.00	1.00	172
Catla	0.85	0.73	0.79	15
Croaker	0.80	0.80	0.80	46
Emperor	0.77	1.00	0.87	10
Indian salmon	0.93	0.79	0.86	145
Lady	0.94	0.92	0.93	92
Mackerel	0.86	0.89	0.88	128
Malabar trevally	0.82	0.56	0.67	16
Mullet	0.84	0.84	0.84	19
Pink perch	1.00	0.98	0.99	127
Rohu	0.60	0.98	0.74	43
Roopchand	0.99	0.97	0.98	100
Sardine	0.99	0.87	0.93	307
Sea bass	0.81	1.00	0.90	13
Seer	0.87	0.99	0.93	163
Tilapia	0.99	0.95	0.97	82
Tuna	0.70	0.82	0.75	28
White pomfret	0.92	0.94	0.93	36
accuracy			0.91	1609
macro avg	0.86	0.89	0.87	1609
weighted avg	0.92	0.91	0.91	1609



7. Future Experiments and Considerations

- Experiment with increased epochs.
- Explore different loss functions.
- Evaluate various dataset sizes.
- Consider utilizing segmented images for improved accuracy.

8. Challenges and Lessons Learned

- Uneven Dataset: Dealing with species with less than 100 images.
- Misclassifications: Tackling issues with Indian salmon, sea bass, sardine, etc.
- Model Tuning: Finding the right balance between model complexity and training duration.

9. Conclusion and Next Steps

7.1 Conclusion:

- The two-stage approach provided promising results, with iterative improvements addressing initial misclassifications.
- The project lays the foundation for a robust species classification model, with ongoing experiments and enhancements to be explored.
- Improved accuracy observed with refined models.

7.2 Next Steps:

- Evaluate model performance on the entire warehouse dataset.
- Implement planned experiments based on results.
- Conduct additional experiments based on findings.